

SDG 13: CLIMATE ACTION

SDG 11: RESILIENT CITIES

SDG 6: CLEAN WATER

JalRakshak

AI-Powered Flood Early-Warning & Advisory Assistant

A conceptual AI early-warning system that combines rainfall intensity with citizen drain-blockage complaint density to generate explainable, ward-level flood risk scores and plain-language advisories in English and Malayalam for urban residents in Kerala.

INTERNSHIP PROJECT

1M1B AI + Sustainability Virtual Internship
Cohort: September 2026

METHODOLOGY & TOOLS

Architecture Planned & Validated via IBM BOB
Rule Engine + Prompt-Based Advisory Layer

TARGET GEOGRAPHY

Kerala Urban Wards
(Kochi Municipal
Corporation)

The Problem: Hyperlocal Early-Warning Gap

Urban flooding in Kerala affects low-lying wards hours before official district alerts arrive

CORE PROBLEM STATEMENT (REQUIRED FORMAT)

"How might we use AI to combine rainfall intensity and local drainage-complaint data so that urban wards in Kerala can get earlier, more localized flood warnings?"

WHO IS AFFECTED

- Low-lying ward residents: Ground floor homes, street stalls, and shops flood repeatedly during routine monsoons.
- Vulnerable citizens: Elderly and marginalized households lacking real-time access to official disaster channels.
- Municipal responders: Ward councillors and health inspectors needing granular, street-level risk visibility.

WHY PROBLEM PERSISTS

- Alerts too broad: Official IMD/KSDMA warnings are district-wide; residents ignore them because they lack street specificity.
- Siloed civic data: Citizens report clogged drains to municipalities (via K-SMART), but this data is never linked to flood models.
- No lead time: Water enters homes before official warnings translate to community action.

CURRENT GAPS TODAY

- No weak-signal fusion: No existing municipal tool combines rainfall forecasts with localized infrastructure bottlenecks.
- Technical jargon: Dam water levels and mm metrics confuse citizens; action-oriented advisories are missing.
- Ignored dry-day risk: Severe drain blockages go uninspected until torrential rainfall triggers catastrophic overflow.

Data Sources: Bridging Meteorology & Civic Infrastructure

Fusing open meteorological data with municipal citizen grievance records

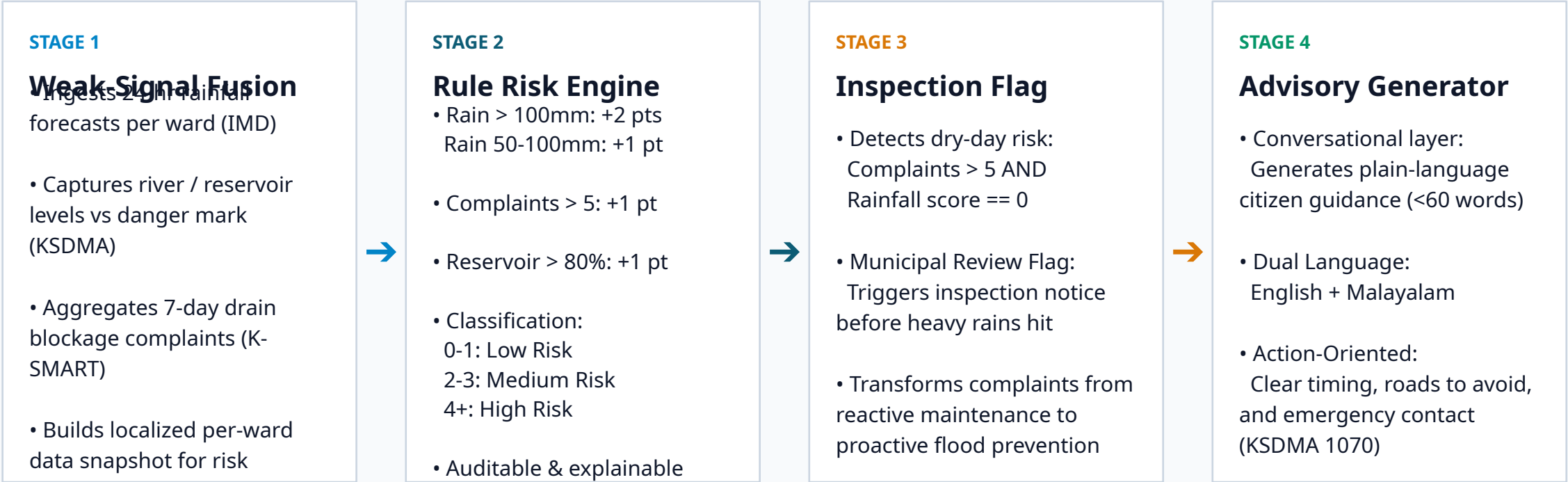
1. RAINFALL INTENSITY	2. RIVER / RESERVOIR LEVELS	3. DRAIN COMPLAINT DENSITY
Real Source: India Meteorological Dept (IMD) • 24-hr rainfall forecasts (mm) • 5-day district-level outlooks & nowcasts • Flash flood bulletins & CAP RSS feeds • Official API: api.imd.gov.in Role in JalRakshak: Primary hazard driver. Identifies incoming water volume across urban catchment basins.	Real Source: Kerala State Disaster Management • Daily reservoir & river level bulletins • % of danger mark (Periyar / Canal gauges) • Early warning river monitoring alerts • Official portal: sdma.kerala.gov.in Role in JalRakshak: System saturation signal. High river levels prevent urban stormwater from discharging outward.	Real Source: K-SMART Municipal Platform (LSGD) • Citizen grievance redressal (launched 2024) • Geo-tagged drain blockage complaints • 7-day unresolved complaint counts per ward • Official portal: lsg.kerala.gov.in Role in JalRakshak: Weak-signal proxy. Blocked drains mean low drainage capacity — transforming moderate rain into flooding.

PROTOTYPE SIMULATION SCOPE (HONEST ENGINEERING BOUNDARY)

For this internship conceptual MVP, live external APIs are intentionally simulated using 4 structured CSV files (wards.csv, rainfall.csv, complaints.csv, river_levels.csv). This isolates and proves the AI scoring and advisory workflow without requiring government credentials.

AI Workflow Architecture: From Raw Signal to Action

Four-stage pipeline converting disparate inputs into localized citizen advisories



WHY RULE-BASED SCORING + PROMPT-BASED ADVISORY (DUAL ARCHITECTURE)

Public disaster management requires 100% auditability — a black-box neural net cannot be explained during emergency council meetings. We keep the risk scoring strictly rule-based, and reserve the AI / LLM layer for language generation and summarizing unstructured citizen complaints.

Explainable Risk Engine & Scoring Rules

Defensible, transparent scoring that links meteorological load to drainage capacity

EXACT POINT ALLOCATION RULES

Input Signal	Threshold Condition	Points	Mechanism
Rainfall (24h)	> 100 mm	+2	Torrents exceed pipe conveyance
Rainfall (24h)	50 mm – 100 mm	+1	Sustained moderate storm run-off
Drain Complaints	> 5 reports (7 days)	+1	Active bottlenecks reduce throughput
River / Reservoir	> 80% danger mark	+1	Backwater effect prevents discharge

Key engineering detail: Strict inequality thresholds (>100mm, >5 complaints, >80% reservoir) prevent boundary jitter. Maximum possible risk score is 4 with current weights.

RISK CLASSIFICATION BANDS

0 – 1 POINTS · LOW RISK

Routine precautions. Drainage network operating within safe margins. Citizen reassurance advisory.

2 – 3 POINTS · MEDIUM RISK

Heightened readiness. Localized waterlogging anticipated on low roads. Move vehicles & secure valuables.

4+ POINTS · HIGH RISK

Immediate action. Severe inundation likely. Avoid low-lying travel after 6 PM. Emergency helpline (KSDMA 1070) broadcast to all registered ward phones.

Prototype Demo: Worked Scenarios A & B

Verifiable outputs from the running JalRakshak Python engine

SCENARIO A: WARD 12 (KALOOR NORTH) — HIGH RISK

Input Signals:

- 24-hr Rainfall: 130 mm (>100mm) → +2 pts
- Drain Complaints (7d): 7 (>5) → +1 pt
- Reservoir Level: 90% (>80%) → +1 pt

Total Score: 2 + 1 + 1 = 4 → HIGH RISK

English Advisory Output:

"High flood risk in Ward 12. Heavy rainfall (130 mm in 24 h); 7 unresolved drain complaints in 7 days; Reservoir/river at 90% of danger mark. Avoid travel through low-lying areas after 6 PM. Keep emergency numbers ready (KSDMA Helpline: 1070)."

Malayalam Advisory Output (Local Language):

"Ward 12-ൽ കനത്ത വെള്ളപ്പൊക്ക സാധ്യതയുണ്ട്. കനത്ത മഴയും അഴുക്കുചാൽ തടസ്സവും സാധ്യത വർദ്ധിപ്പിക്കുന്നു. താഴ്ന്ന പ്രദേശങ്ങളിലൂടെയുള്ള യാത്ര ഒഴിവാക്കുക. അടിയന്തര സഹായത്തിന് 1070 നമ്പറിൽ ബന്ധപ്പെടുക."

SCENARIO B: WARD 5 (PANAMPILLY NAGAR) — LOW RISK

Input Signals:

- 24-hr Rainfall: 60 mm (50-100mm) → +1 pt
- Drain Complaints (7d): 2 (<=5) → +0 pts
- Reservoir Level: 60% (<=80%) → +0 pts

Total Score: 1 + 0 + 0 = 1 → LOW RISK

English Advisory Output:

"Low flood risk in Ward 5 today. Normal precautions advised."

Malayalam Advisory Output (Local Language):

"Ward 5-ൽ ഇന്ന് വെള്ളപ്പൊക്ക ഭീഷണി കുറവാണ്. സാധാരണ മുൻകരുതലുകൾ തുടരുക."

Insight Beyond Prediction: Scenario C

Detecting infrastructure failure before rainfall arrives — transforming complaints to proactive maintenance

SCENARIO C: WARD 9 (PONEKKARA) — LOW WEATHER RISK BUT MUNICIPAL INSPECTION FLAGGED

INPUT SIGNALS & ENGINE EVALUATION

- 24-hr Rainfall: 40 mm (<50mm) → +0 pts (dry day signal)
- Drain Complaints (7d): 8 reports (>5) → +1 pt (bottleneck)
- Reservoir Level: 70% (<=80%) → +0 pts (safe capacity)

Total Score: 0 + 1 + 0 = 1 → Classified as LOW RISK

🚩 INFRASTRUCTURE REVIEW RULE FIRED:
comp_pts > 0 AND rain_pts == 0

Generated Advisories (Municipal Action + Citizen):

EN: "Low rainfall risk, but Ward 9 has 8 unresolved drain complaints — recommend municipal inspection regardless of rainfall."

ML: "മഴ കുറവാണ്, എങ്കിലും Ward 9-ൽ 8 അഴുക്കുചാൽ പരാതികൾ പരിഹരിക്കപ്പെടാതെ കിടക്കുന്നു — അടിയന്തര നഗരസഭാ പരിശോധന ശുപാർശ ചെയ്യുന്നു."

WHY THIS IS A BREAKTHROUGH FOR CITY RESILIENCE

1. Beyond Pure Weather Prediction:
Standard flood apps only trigger alerts when skies open up. JalRakshak identifies the root vulnerability — choked drainage — before the clouds arrive.
2. Proactive Municipal Desilting:
Ward health inspectors receive an automated priority list of drains that must be cleared within 48 hours, turning civic complaints into an engineering defense line.
3. Citizen Accountability Loop:
Citizens who submitted complaints see their reports actively monitored as early warning indicators, increasing trust in municipal platforms like K-SMART.

Planned & Validated with IBM BOB

Using IBM BOB Ask/Plan mode to reason through architecture and stress-test scaling limits

KEY GAPS IDENTIFIED BY IBM BOB

- 1. Temporal Misalignment at Scale (6 → 100+ Wards):**
BOB flagged that static daily snapshots fail when IMD nowcasts update 3-hourly while complaint logs stream continuously. Recommended asynchronous event-driven queues.
- 2. Spatial Gauge Granularity:**
Current prototype applies city-wide max reservoir level to all wards equally. BOB flagged that inland wards should not inherit coastal/river gauge alerts — mandates spatial basin mapping.
- 3. Complaint Normalization Bias:**
Flat complaint count (>5) penalizes smaller wards and ignores dense wards. BOB recommended per-capita weighting (complaints / 1,000 households).
- 4. LLM Hallucination Guardrails:**
BOB warned that LLMs confabulate street names in Kerala. We adopted deterministic templates as primary runtime and added KSDMA Helpline 1070.

SCREENSHOT 1: IBM BOB ARCHITECTURE REASONING

[Drop Screenshot of IBM BOB Ask/Plan Mode here]

Shows: Initial prompt reasoning through ward-level risk scoring, data flow architecture, and weak-signal fusion.

SCREENSHOT 2: IBM BOB GAP ANALYSIS & CRITIQUE

[Drop Screenshot of IBM BOB Gap Analysis here]

Shows: BOB flagging scaling risks, unnormalized complaints, and recommending emergency helpline validation.

Engineering Scope: Prototype vs Production Scaling

Honest engineering boundaries — what is built today versus roadmap for full deployment

IN SCOPE: DELIVERED MVP PROTOTYPE

✓ Explainable Risk Engine:

Rule-based scoring (+2, +1, +1, +1) implemented in clean, testable Python.

✓ Multi-Source Simulated Dataset:

6 Kochi wards with rainfall, river gauges, and 7-day complaint logs.

✓ Dual-Language Advisory Generator:

Deterministic template engine producing English and Malayalam citizen SMS.

✓ Scenario C Infrastructure Detection:

Automated municipal inspection trigger for high-complaint dry days.

✓ 100% Automated Test Suite:

Unit test suite (unittest) verifying boundary conditions and spec compliance.

✓ IBM BOB Validation Artifacts:

Comprehensive architecture critique (.bob rules & markdown review).

OUT OF SCOPE: PRODUCTION-SCALING ROADMAP

→ Live Government API Connections:

Real-time data feeds with IMD Mausam API and KSDMA alert webhooks.

→ Spatial Hydrological DEM Modeling:

Digital Elevation Models (DEM) mapping slope run-off directly into drains.

→ Population-Normalized Complaints:

Weighing complaints by ward household density and road network length.

→ Officer-in-the-Loop Broadcast Portal:

Web dashboard for municipal emergency officers to approve High alerts.

→ Automated Telephony & Voice SMS:

Twilio / Kerala BSNL cell-broadcast integration for Malayalam voice IVR.

→ Historical Flood Calibration:

Calibrating score weights against 2018–2024 historical flood inundation maps.

Building Climate-Resilient Urban Communities

How JalRakshak delivers direct progress toward the United Nations Sustainable Development Goals

PRIMARY SDG

SDG 13: Climate Action

Target 13.1 & 13.3:
Strengthen resilience and adaptive capacity to climate-related hazards.

Impact:
Provides localized early warning 3–6 hours before inundation, allowing families in low-lying wards to secure belongings, avoid hazardous roads, and prevent loss of life.

SECONDARY SDG

SDG 11: Resilient Cities

Target 11.5 & 11.B:
Reduce direct disaster losses and strengthen integrated disaster risk management in cities.

Impact:
Shifts municipal drainage maintenance from reactive complaints to predictive flood prevention. Prioritizes ward cleaning before monsoon deluges strike.

SECONDARY SDG

SDG 6: Clean Sanitation

Target 6.3:
Improve water quality by eliminating dumping and minimizing release of pollutants.

Impact:
Preventing urban drain stagnation stops blackwater backflow into wells and ground contamination, curbing waterborne disease epidemics like cholera and leptospirosis.

JalRakshak demonstrates how weak civic signals, fused with AI, create life-saving resilience for urban India.